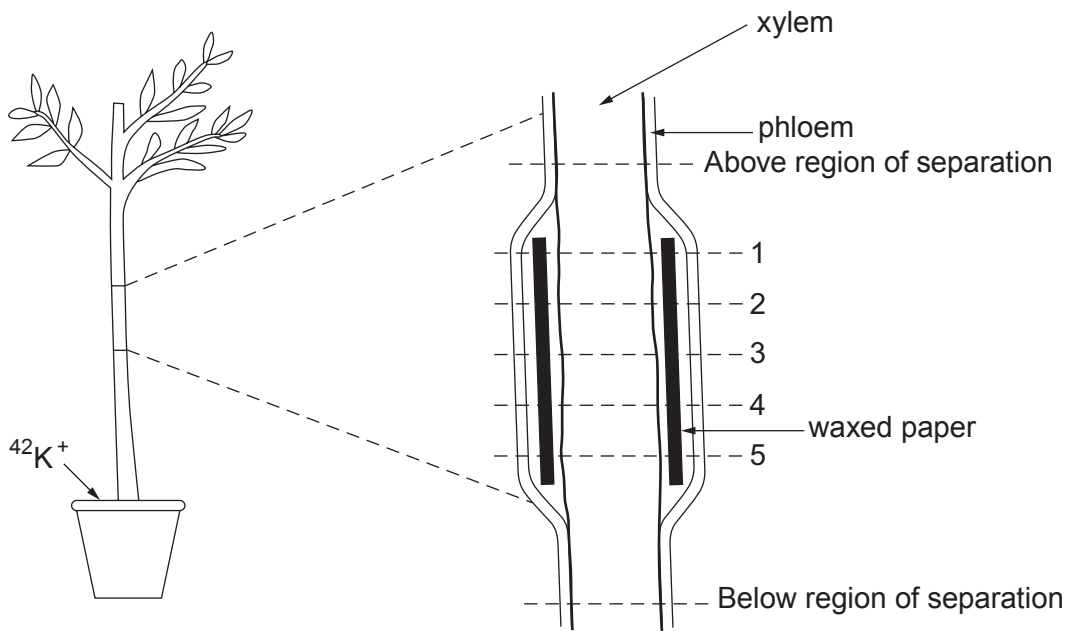


6. (a) The apparatus below was set up to study the upward transport of mineral ions from the soil. The mineral ion used was radioactive potassium,  $^{42}\text{K}^+$ .

In one region of the stem, the xylem was separated from the phloem and waxed paper was inserted between the xylem and phloem to act as a barrier between the tissues. This is illustrated in the simplified diagram below.



The plants were allowed to absorb  $^{42}\text{K}^+$  for five hours. After this period, the distribution of  $^{42}\text{K}^+$  in the xylem and phloem at points above and below the region of separation, and at points 1 to 5 within the region of separation, were measured. The results are shown in the table.

| Sampling point | Radioactivity (ppm $^{42}\text{K}^+$ ) |       |
|----------------|----------------------------------------|-------|
|                | phloem                                 | xylem |
| Above          | 53                                     | 55    |
| 1              | 11                                     | 119   |
| 2              | 0.3                                    | 122   |
| 3              | 0.4                                    | 112   |
| 4              | 0.3                                    | 110   |
| 5              | 0.3                                    | 108   |
| below          | 56                                     | 59    |



- (i) What conclusion can be made from these results about the transport of  $^{42}\text{K}^+$  in the xylem and phloem? [2]

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- (ii) A suitable control for this experiment would be to separate the xylem and phloem but not to insert waxed paper.

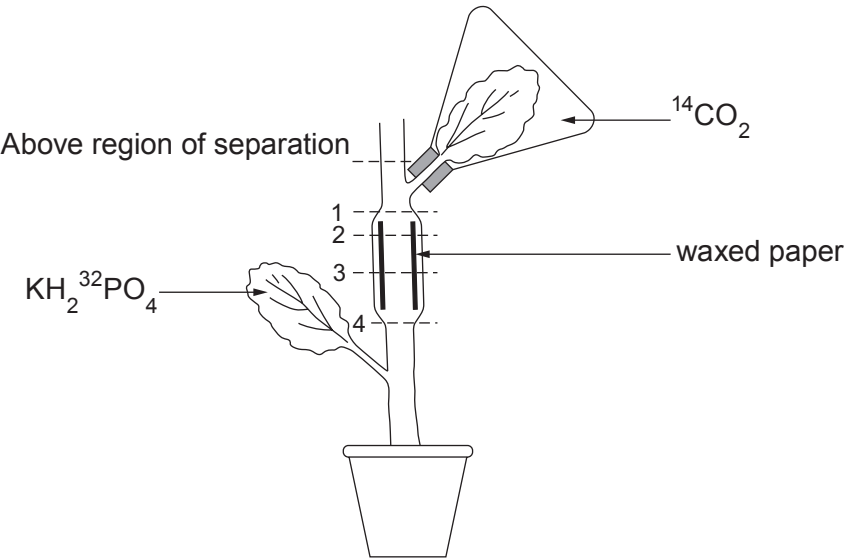
Suggest why this would be a suitable control and explain how the results could be used to support your conclusion. [2]

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- (b) In a similar experiment  $^{14}\text{CO}_2$  was applied to an upper leaf and  $\text{KH}_2^{32}\text{PO}_4$  to a lower leaf. After fifteen hours, the levels of both radioactive isotopes  $^{14}\text{C}$  and  $^{32}\text{P}$  were analysed. The results are shown in the table.



| Sample site | Radioactivity (ppm) |       |                 |       |
|-------------|---------------------|-------|-----------------|-------|
|             | $^{14}\text{C}$     |       | $^{32}\text{P}$ |       |
|             | phloem              | xylem | phloem          | xylem |
| Above       | 6 500               | 0     | 98              | 0     |
| 1           | 3 480               | 0     | 103             | 0     |
| 2           | 3 030               | 0     | 116             | 0     |
| 3           | 2 380               | 0     | 125             | 0     |
| 4           | 2 300               | 0     | 185             | 0     |

- (i) What conclusions can be made from these results about transport of  $^{14}\text{C}$  in xylem and phloem? [2]

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- (ii) Describe how you would refine the experiment to determine the direction of transport of  $^{32}\text{P}$ . [2]

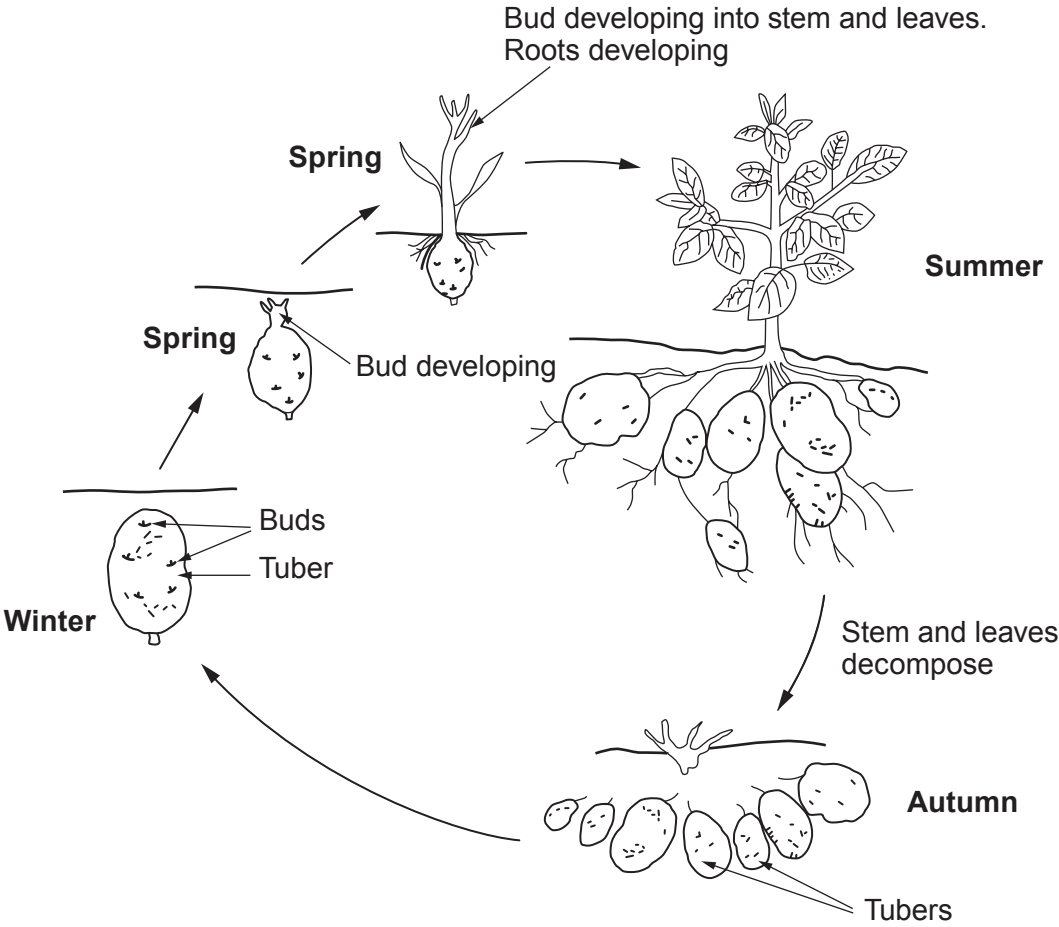
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(c) Organic molecules such as sucrose move through a plant in the phloem by mass flow.



The diagram shows the life-cycle of a potato plant. During the summer months the potato tuber acts as a sink and stores starch. In Spring the tuber acts as a source and supplies sugars to the new shoots and roots. Use the mass flow model to explain how the sugars are transported to the potato tuber to be stored as starch and explain the advantage of the tuber acting as a source in Spring. [5]

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